COURSE

Cambridge IGCSE Business (0450 / 0264)

Break-even

4 Operations management · 4.4 Costs, scale of production and break-even analysis (0450: 4.2)

Revision for 4 Operations management: each firm works it out on its own deals.

Class **Cambridge IGCSE Business (0450 / 0264)** Topic **Break-even**

THE LESSON

Setup	2 min

Round 1 — Decisions 3 min · Deals 5 min · Topic tasks 4 min

Firm decisions	**Deals**	TASK 1 **Cars to break even**	TASK 2 **Lower break-even**

Round 2 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	**Deals**	TASK 3 **Rank by break-even**	TASK 4 **Profit on the deal**	**B2B deal**

Round 3 — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 2 min

Firm decisions	**Deals**	TASK 5 **Less risky deal**	TASK 6 **New break-even**	**B2B deal**

Final questions	3 min

45-minute setting shown. The 30-minute setting has one topic task per round (Tasks 3, 5 and 6) and one B2B deal. 60 and 75 add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments	Buy property or funds. The value can rise or fall.
Operations	**Buying**	Hire the Procurement Specialist or the Stock Controller for the round.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

They take up to 8 cars.

Cars to break even?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Livery and set-up, one-off	**£8,750**
Customer pays per car	**£9,750**
You pay per car	**£8,000**

METHOD

Break-even = set-up ÷ (customer pays – you pay), per car

£9,750 − £8,000 = £1,750; £8,750 ÷ £1,750 = 5

Why: Each car adds £1,750, so 5 cars pay the set-up.

Task 2

ROUND 1

You want to lower your break-even point.

Which two changes would lower it?

PICK TWO

- ✔ **Cut the fixed costs**
- ○ Pay higher rent
- ○ Pay more for each car
- ✔ **Raise the selling price**
- ○ Sell more cars

Why: More added per car or lower fixed costs; selling more does not move it.

Task 3

ROUND 2

Three deals: fixed costs, and what each car adds (price minus cost).

Order by break-even, fewest cars first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **£6,000 fixed, £1,500 a car**
2. **£9,000 fixed, £1,500 a car**
3. **£4,000 fixed, £500 a car**

Why: Fixed costs ÷ what each car adds: 4, 6 and 8 cars.

Task 4

ROUND 2

A deal with a set-up cost.

Profit on the deal?

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Livery and set-up, one-off	**£7,000**
Customer pays per car	**£10,750**
You pay per car	**£9,000**
They order	**9 cars**

METHOD

Profit = cars ordered × (customer pays − you pay) − set-up

9 × (£10,750 − £9,000) − £7,000 = £8,750

Why: The first cars pay the set-up; the rest is profit: £8,750.

Task 5

ROUND 3

Both deals are for 10 cars. The taxi deal breaks even at 4 cars, the hire deal at 8.

Which deal is less risky, and why?

PICK A OR B AND GIVE A REASON

- ○ The hire deal
- ✔ **The taxi deal**

Why: The taxi deal's margin of safety is 6 cars; the hire deal's only 2.

Task 6

ROUND 3

They want £400 off a car.

New break-even in cars? Round up.

WORK OUT THE NUMBER · EXAMPLE FIGURES

Livery and set-up, one-off	**£5,500**
Customer pays per car	**£19,250**
You pay per car	**£17,000**
Price cut per car	**£400**

METHOD

Break-even = set-up ÷ (new price − you pay), rounded up

£18,850 − £17,000 = £1,850; £5,500 ÷ £1,850 → 3

Why: Less added per car (price minus cost) means more cars to break even.

Final questions

6
questions for each pupil

159
Level 2 Business questions

0
of them on break-even

No question in the pool is about break-even yet. The game draws 6 of the 159 Level 2 questions.